What Is a Good Model?

Provost & Fawcett, Data Science for Business, chapters 7–8

AI in Business · Week 3

Last week's cliffhanger, and today's plan

74%

of the 200 customers ranked highest by our hand-made score really churned, against 26.5% on average.

But should we call them? How many?

1

What is a good model?

Ch. 7: accuracy, confusion matrix, expected value, baselines

45 min

2

Seeing model performance

Ch. 8: ranking, profit, lift and ROC curves

40 min

1. What is a good model?

Chapter 7: decision-analytic thinking

Accuracy can look good and mean nothing!

99.9%

accuracy from always predicting "no" when classes are 999 : 1

90%

accuracy from "nobody churns" when 10% of customers churn (MegaTelCo)

73.5%

accuracy from "nobody churns" on our Telco data. Useless, yet it sounds fine

Always compare with the base rate. Accuracy = correct decisions / all decisions. With unbalanced classes, predicting the majority class already scores high.

The confusion matrix: four kinds of decisions

Predicted Class	Actual Class: Positive	Actual Class: Negative
Churn	True positive	False positive
Not Churn	False negative	True negative

In churn terms (Jupyter Notebook)

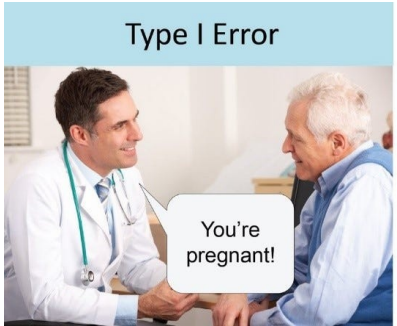
True positive: we predicted churn, and they left

False positive: we predicted churn, but they stayed (Type I Error)

False negative: we said they would stay, but they left (Type II Error)

True negative: we said they would stay, and they did

Note: In statistics and ML, "positive" means worth our attention (eg. death is positive, living is 0/negative)
In business cases, Positive means churn, fraud, disease, not necessary "good" in sociological terms!





Same accuracy, very different models

Model A



Model B



Balanced test set (50 : 50)	**A: 80%**	**B: 80%**
Real population (10% churn)	**A: 64%**	**B: 96%**

Which is better? We cannot say without costs, and context!

Example from ch. 7: 1,000 customers. A catches every churner but raises many false alarms; B never raises a false alarm but misses most churners.

Errors do not cost the same (Cost-Benefit)



Medical test

False positive

Told she has cancer, but does not: more tests, stress, expense.

False negative

Told she is healthy, but has cancer: missed early treatment.



Churn

False positive

Retention offer to someone who stays anyway: we pay for nothing.

False negative

No offer to someone who leaves: we lose a customer.

The general principle: When we are applying data science to an actual application it is vital to return to the question: what is important in the application? What is the goal? Are we assessing the results of data mining appropriately given the actual goal?

Expected value: the framework for the whole course

$$EV = p(o_1) \cdot v(o_1) + p(o_2) \cdot v(o_2) + p(o_3) \cdot v(o_3) + \dots$$

Probability of each outcome × its value, added up: a dot product, as in week 2.



Structure

Which outcomes are possible? Lay out the decision.



Probabilities

Usually estimated from data, by a model.



Values

Usually from the business: costs, margins, experts.

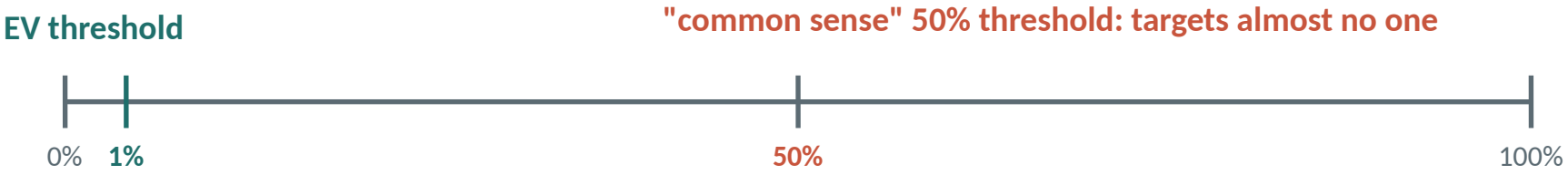
Should we target this customer? A threshold from EV

A product sells for $200 and costs $100 to make. Mailing an offer costs $1.

If she buys
+$99

If she does not
−$1

Target if $p \times 99 - (1 - p) \times 1 > 0$
→ p > 1%



Response rates are often 1–2%. With a 50% cut-off, nobody is ever a "likely responder".

Bottom Line: Understanding Problem is Crucial → Domain Expertise + Machine Learning!

Activity 1: Who gets the offer?



In pairs · 8 minutes · calculator allowed

An offer costs €2 to send. A customer who responds brings €38 profit after that cost; one who does not respond costs us the €2.

Lali
p(respond) = 2%

Fiku
p(respond) = 4%

Noci
p(respond) = 8%

Rroku
p(respond) = 30%

1. Compute the expected value of targeting each customer.
2. What is the break-even probability?

A whole model's value: confusion matrix × benefits

Test-set counts (110 consumers)

	actual p	actual n
Y	56	7
N	5	42

×

Benefit of each decision ($)

	actual p	actual n
Y	+99	−1
N	0	0

=

≈ $50
expected profit per consumer

$$56/110 \times \$99 \;+\; 7/110 \times (-\$1) \;+\; 0 \;+\; 0$$

Divide each count by the total to get a probability, multiply by its value, add up. Now models can be compared in money, on the same population.

Two easy mistakes in cost-benefit matrices

Inconsistent signs

Pick one view and keep it: benefits positive and costs negative (as in the book), or the reverse. Never mix the two in one matrix.

Double counting

Fraud costs $1,000. Writing +$1,000 for catching it and −$1,000 for missing it makes one catch worth $2,000. Put the value in one cell; the other is zero.

Quick test: change one decision and ask how much better off you are. If the answer is too big, you double counted.

Always beat a sensible baseline, not just random

Majority class

Predict the most common class. 94% accuracy is nothing if 94% are negatives.

Simple rules

Weather: "same as today" and "historical average" are hard to beat.

One feature

A decision stump: one informative variable. Often surprisingly good.

Received wisdom

What experts already do: e.g. go after the biggest past customers.

Our baseline in the lab: last week's hand-made risk score. Does a real model beat it?

2. Seeing model performance

Chapter 8: profit, lift and ROC curves

Rank customers, then choose where to cut

Score	Really churned?	
0.91	yes (p)	
0.84	yes (p)	
0.77	no (n)	
		Cut A: top 3
0.70	yes (p)	
0.62	no (n)	
0.55	yes (p)	
		Cut B: top 6
0.41	no (n)	
0.33	no (n)	
0.24	yes (p)	
0.12	no (n)	

Each cut gives a different confusion matrix

Cut A

	actual p	actual n
Y	2	1
N	3	4

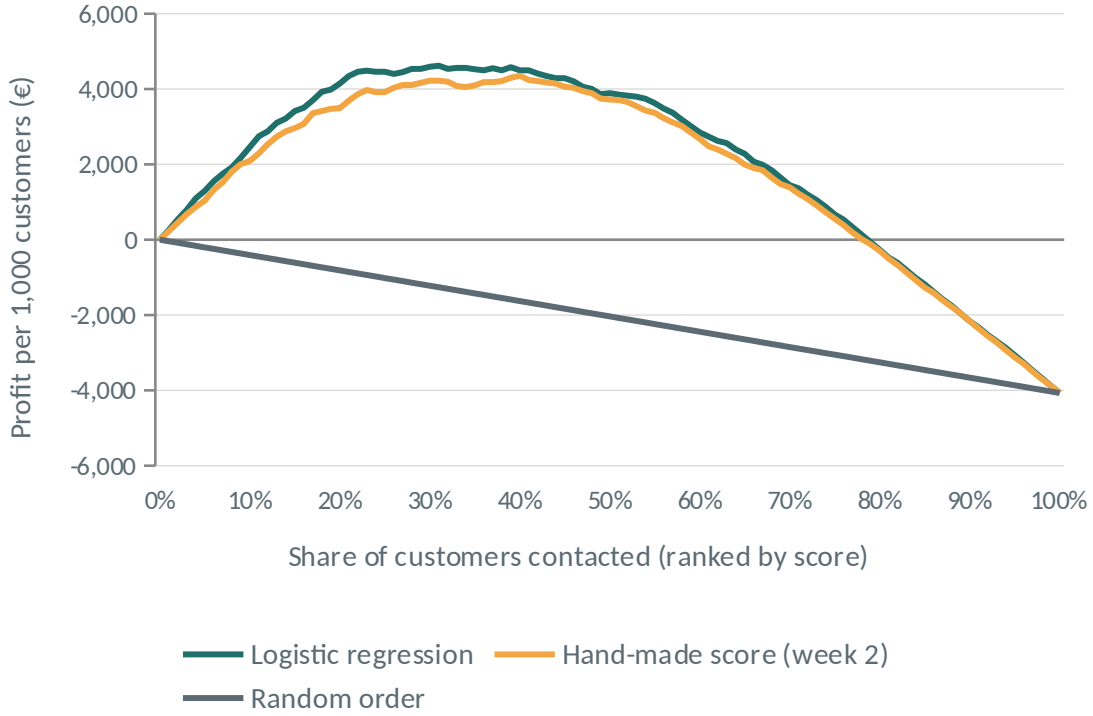
Cut B

	actual p	actual n
Y	4	2
N	1	3

Why rank instead of classify?

- A fixed budget: call the top N
- Scores may rank well even if they are not exact probabilities
- Costs may be uncertain: see the curves next

Profit curve: how far down the list should we go?



Assumptions

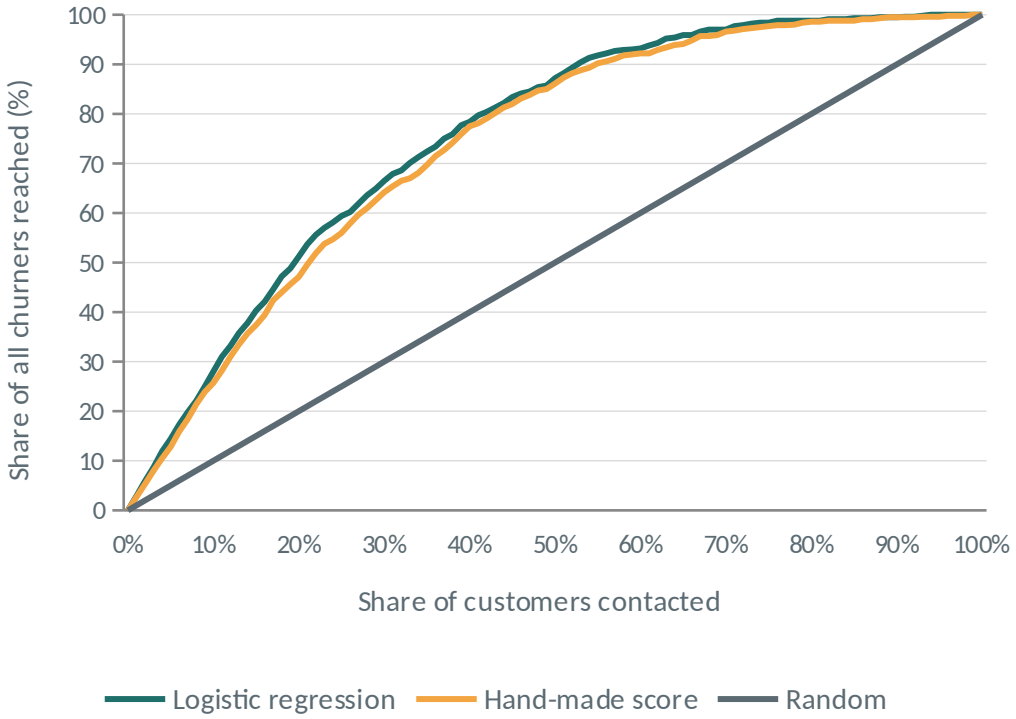
A call plus discount costs €20. A contacted churner is kept 1 time in 4, worth €240 → a true positive is worth +€40; a false positive −€20.

Best: contact the top 31%
€4,619 per 1,000 customers

Contact everyone
−€4,070 per 1,000 customers

Telco test set (2,113 customers, churn 26.5%). Assumptions are illustrative; you will change them in the lab.

Cumulative response and lift: no costs needed

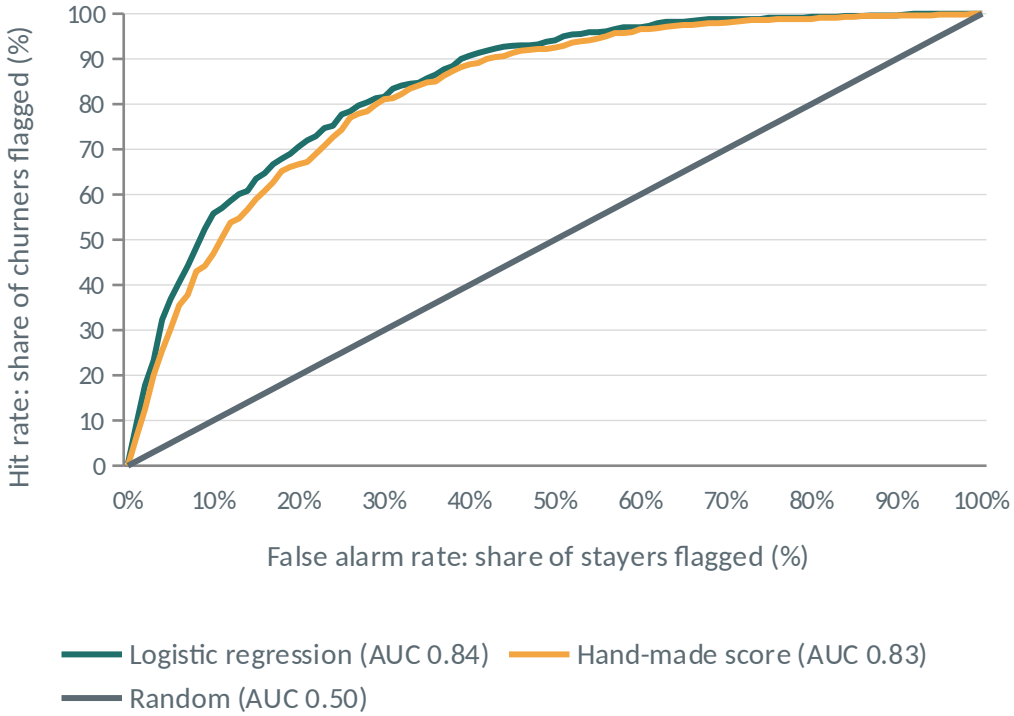


2.8× lift

The top 10% of the list holds 28% of all churners. Random would hold 10%.

- Lift = model's hit rate ÷ random hit rate at the same point
- Easy to explain to managers
- Assumes the test set has the real share of churners

ROC and AUC: ranking quality apart from costs



- Each point is one cut-off: hit rate vs. false alarm rate
- Top-left corner = perfect. Diagonal = guessing
- Does not change with costs or the share of churners

AUC

Area under the curve, 0.5 to 1. The chance that a random churner is ranked above a random stayer.

Which chart when?



Profit curve

Use when: Costs, benefits and class mix are known and stable

Best for managers: shows money



Cumulative response / lift

Use when: Class mix known, costs uncertain

Easy to explain: "top 10% holds 28% of churners"



ROC and AUC

Use when: Costs and class mix uncertain or changing

For analysts comparing models

Rule of thumb: show managers profit or lift; keep ROC and AUC for comparing models.

The book's churn case: accuracy points the wrong way

Model	Accuracy on training data	Accuracy (cross-validated)	AUC (cross-validated)
Classification tree	95%	91.8%	0.614
Logistic regression	93%	93.0%	0.574
k-nearest neighbour	**100%**	93.0%	**0.537**
Naive Bayes	76%	**76.5%**	**0.632**

- 100% on training data = memorising, not learning (overfitting).
- k-NN's 93% just copies the base rate: it almost never predicts churn.
- Naive Bayes has the lowest accuracy but ranks churners best (highest AUC).

KDD Cup 2009, Orange (France Telecom): about 47,000 customers, 7% churn. Ch. 8, tables 8-1 and 8-2.

Activity 2: Read the profit curve



In pairs · 7 minutes · use the profit curve (slide 16)

1. No budget limit: which ranking would you use, and how far down the list would you go?
2. Budget for only 10% of customers: which ranking, and what profit do you expect?
3. Marketing proposes calling every customer "to be safe". What do you answer?
4. The offer turns out to cost €30, not €20. Which way does the best cut-off move, and why?

Five things to remember

1. Accuracy hides what matters: always check the base rate
2. Count each kind of error separately; they cost different amounts
3. Expected value = probabilities from data × values from the business
4. The best threshold comes from costs, not from 50%
5. Pick the chart for your uncertainty: profit, lift or ROC

After the break: build these curves yourself on the Telco data